



WEEK 07

Internship Report



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Contents

1. Introduction	2
2. Duration.....	2
Week 07:.....	2
3. Overview.....	2
4. Tasks Performed	2
4.1 Core UI Screens & Navigation	2
4.2 Player Implementation	3
4.3 Media Management Features.....	3
A. Playlist Management.....	3
B. Duplicate Detection & Management	3
C. Media Tag Editor	3
D. Folder Browser	3
4.4 Advanced Features.....	4
5. Learning Outcomes	5
5.1 Technical Skills Acquired	5
5.2 Key Understandings.....	5
6. Challenges and Solutions	5
Challenge 1:.....	5
Challenge 2:.....	6
Challenge 3:.....	6
Challenge 4:.....	6
Challenge 5.....	6
7. Conclusion	6

1. Introduction

Building upon the robust foundation established in Week 06, Week 07 focuses on implementing the user interface, advanced media management features, and optimizing the overall user experience.

This phase transforms the core backend services into a fully functional media player with intuitive controls, comprehensive media library management, and advanced capabilities such as equalizer support, floating picture-in-picture, and duplicate detection.

2. Duration

Week 07: UI/UX implementation, advanced feature development, and optimization phase.

3. Overview

The Week 07 deliverables of NYX Media Player include all user-facing components, advanced media handling, and performance optimizations. The application now provides:

- I. User Interface: 10+ interactive screens and views
- II. Media Library Management: Browsing, searching, and organizing media
- III. Advanced Playback Controls: Equalizer, gestures, floating player
- IV. Media Intelligence: Duplicate detection, tag editing, statistics
- V. Platform Features: Share sheet, Picture-in-Picture (PIP), edge panel
- VI. User Preferences: Gesture customization and profile management

4. Tasks Performed

4.1 Core UI Screens & Navigation

- Shell View (main container with bottom navigation)
- Onboarding View (first-time setup and permissions)
- Home Screen (dashboard with quick access)

Media Library Views:

- Audio/Music browser

- Video browser
- Playlist management
- Search Screen (full-text search with filters)
- Settings Screen (user preferences and configuration)

4.2 Player Implementation

- Full-screen Player View
- Playback controls (play, pause, seek, skip, shuffle, repeat)
- Media visualization with metadata and artwork
- Mini Player for background usage
- Floating Player (Picture-in-Picture mode)
- In-app PIP overlay (floating orb)

4.3 Media Management Features

A. Playlist Management

- Create, edit, and delete playlists
- Add/remove media items
- Drag-and-drop reordering
- Persistent storage

B. Duplicate Detection & Management

- Detect duplicates via hash/similarity
- Batch removal functionality
- Smart grouping of duplicates
- Metadata-preserving merge

C. Media Tag Editor

- Edit ID3 tags (Title, Artist, Album, etc.)
- Cover art modification
- Batch editing support
- File synchronization

D. Folder Browser

- Directory-based browsing
- Folder statistics
- Nested navigation

- Quick folder access

4.4 Advanced Features

A. Equalizer System

- Multi-band equalizer
- Preset management
- Custom EQ profiles
- Real-time audio adjustments
- Bass and treble controls

B. Gesture Recognition

- Custom gesture controls
- Volume and brightness adjustments
- Playback gestures
- Persistent user preferences

C. Floating Panel System

- Edge panel for quick controls
- Playlist access panel
- Floating buttons
- Gesture-based interaction

D. Background Playback Management

- Continuous background audio playback
- Notification-based controls
- Foreground service integration
- Battery optimization

4.5 Media Discovery & Organization

A. Search Functionality

- Full-text search (title, artist, album)
- Filters (type, duration, date)
- Search history
- Auto-suggestions

B. Media Store Integration

- System media access
- Device library synchronization

- Real-time updates on file changes
- Indexed performance optimization

C. Thumbnail Generation

- Video thumbnail creation
- Cached thumbnails for performance
- Artwork extraction
- Album art display

5. Learning Outcomes

5.1 Technical Skills Acquired

- Flutter UI Development
- Responsive layouts
- Custom widgets
- Animations and transitions
- Material Design 3
- Accessibility practices
- State Management (Provider)
- ViewModel patterns
- Observer pattern
- Hierarchical state management
- Selective rebuild optimization
- Advanced Media Processing

5.2 Key Understandings

- Balancing feature richness with simplicity is critical
- Gesture controls enhance UX but require feedback
- Background playback demands proper lifecycle handling
- Caching improves performance significantly
- Duplicate detection is complex but essential

6. Challenges and Solutions

Challenge 1: Complex State with Multiple Services

Solution:

Implemented ViewModels with Provider to ensure clear separation of concerns.

Challenge 2: Floating Player Implementation

Solution:

Developed an in-app PIP overlay widget with manual positioning and drag handling.

Challenge 3: Duplicate Detection Performance

Solution:

Utilized isolates for hash computation to avoid UI blocking.

Challenge 4: Thumbnail Generation for Videos

Solution:

Implemented an asynchronous thumbnail service with caching.

Challenge 5: Metadata Editing Persistence

Solution:

Used database-backed tag storage with synchronization to media files.

7. Conclusion

Week 07 successfully transformed NYX Media Player from a backend-focused system into a complete, feature-rich media application.

Key Achievements:

- a. Complete UI with 10+ screens
- b. Advanced media management system
- c. Feature-rich playback (EQ, gestures, PIP)
- d. Intelligent media processing
- e. Optimized performance and caching
- f. Professional UX design